

Automation Exercise – Capstone Project Overview

1. Project Introduction

This project is an end-to-end automation testing framework built using the Automation Exercise website. It focuses on validating core e-commerce functionalities through UI and API automation testing.

The framework is designed using the Page Object Model (POM) with Playwright and JavaScript to ensure scalability, maintainability, and readability.

2. Project Objective

The main objective of this project is to:

- Automate key user flows of an e-commerce application
 - Validate functional correctness of UI and API layers
 - Ensure reliability of critical business operations
 - Demonstrate professional automation testing structure
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3. Website Under Test

- **Name:** Automation Exercise
 - **Type:** E-commerce Testing Website
 - **Use Case:** Practice website for automation testing scenarios
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4. Tools & Technologies Used

- Playwright
 - JavaScript (Node.js)
 - HTML/CSS (for understanding UI flow)
 - Page Object Model (POM)
 - API Testing (Playwright request)
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5. Framework Design Approach

The project follows:

- Page Object Model (POM) design pattern
- Modular test structure

- Reusable components
 - Separation of test logic and page actions
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6. Test Coverage (8 Services)

1. Authentication (15 Test Cases)

- Signup
- Login
- Logout
- Invalid login validation
- Forgot password

2. Product Module (20 Test Cases)

- Product listing
- Search functionality
- Product details page
- Category filtering

3. Cart Module (15 Test Cases)

- Add to cart
- Remove from cart
- Update quantity
- Cart validation

4. Checkout & Order (20 Test Cases)

- Checkout process
- Address confirmation
- Payment flow
- Order placement

5. User Profile (10 Test Cases)

- Profile update
- Password change
- Email update

6. Shipping Address (10 Test Cases)

- Add address
- Edit address
- Delete address
- Field validation

7. Customer Support (10 Test Cases)

- Contact form submission
- Form validation

- Email confirmation

8. API Testing (20 Test Cases)

- GET product APIs
 - POST login API
 - POST signup API
 - Cart API validation
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7. Total Test Case Summary

Module	Test Cases
Authentication	15
Product Module	20
Cart Module	15
Checkout & Order	20
User Profile	10
Shipping Address	10
Customer Support	10
API Testing	20
Total	120

9. Key Features

- Page Object Model (POM) implementation
 - UI + API hybrid automation
 - Data-driven testing approach
 - Parallel test execution support
 - Scalable test framework design
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10. Execution Commands

```
npm install
npx playwright install
npx playwright test
npx playwright show-report
```

11. Conclusion

This automation project demonstrates a complete real-world testing framework covering UI and API testing. It ensures end-to-end validation of an e-commerce application and follows industry-standard automation practices using Playwright and POM architecture.